

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0046

Shroud 100 kV
June 02, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0046. The Interconnection Customer proposes a 514.7 MW natural gas combined cycle facility interconnecting at the Shroud 100 kV substation in the Indiana area. The request was received on June 15, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 37 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,012,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of SHROUD – SOUTHEAST CHICAGO ENERGY PROJECT 100 kV, the ELM CREEK – NACOGDOCHES RD 100 kV Line loads to 193.1% of its 249 MVA rating. Project DFAX: 51.9%.

SHROUD – SOUTHEAST CHICAGO ENERGY PROJECT 100 kV: 186.1% loading (338 MVA) following the outage of the SHROUD – ELM CRK 100 kV line; DFAX = 65.1%.

The HUISKAMP – WILDWOOD SUBSTATION 100 kV is overloaded to 183.3% of its 140 MVA emergency rating for the loss of the SHROUD – SOUTHEAST CHICAGO ENERGY PROJECT 100 kV. The project contributes 28.1% DFAX (144.5 MW) to this overload.

For the loss of SHROUD – SOUTHEAST CHICAGO ENERGY PROJECT 100 kV line, the SHROUD – ELM CRK 100 kV line loads to 176.4% of its 301 MVA rating. Project DFAX: 51.9%.

For the loss of HUISKAMP – HEGEWISCH 100 kV line, the OUTLOOK – HEGEWISCH 100 kV loads to 173.4% of its 122 MVA rating. Project DFAX: 19.7%.

The LAS VEGAS 6306 – WEST MEMPHIS GATEWAY 100 kV is overloaded to 161.5% of its 126 MVA emergency rating for the loss of the SHROUD – SOUTHEAST CHICAGO ENERGY PROJECT 100 kV. The project contributes 17.8% DFAX (91.8 MW) to this overload.

HUISKAMP – HEGEWISCH 100 kV: 157.9% loading (198 MVA) following the outage of the SOUTHEAST CHICAGO ENERGY PROJECT – SHROUD 100 kV; DFAX = 28.1%.

For the loss of JAMESON – FRANK E RATTS STATION 100 kV line, the FLETCHER CHALLENGE – WARRENSBURG 100 kV loads to 137.0% of its 179 MVA rating. Project DFAX: 12.5%.

For the loss of SOUTHEAST CHICAGO ENERGY PROJECT – SHROUD 100 kV, the NACOGDOCHES RD – LAS VEGAS 6306 100 kV loads to 136.9% of its 186 MVA rating. Project DFAX: 17.8%.

For the loss of TOWER AUTOMOTIVE – HEGEWISCH 100 kV Line, the TOWER AUTOMOTIVE – HEGEWISCH 100 kV loads to 136.5% of its 255 MVA rating. Project DFAX: 25.3%.

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The JAMESON – FRANK E RATTS STATION 100 kV Line is overloaded to 135.2% of its 190 MVA emergency rating for the loss of the W MEMPHIS GATEWAY – HUGO 100 kV. The project contributes 14.1% DFAX (72.8 MW) to this overload.

For the loss of FISK STREET – WESTON VILLAGE 230 kV, the W MEMPHIS GATEWAY – HUGO 100 kV loads to 127.9% of its 208 MVA rating. Project DFAX: 16.2%.

HUGO – BLUFFVIEW 100 kV line: 125.9% loading (114 MVA) following the outage of the PEACH ORCHARD – HUGO 100 kV; DFAX = 13.1%.

The FLETCHER CHALLENGE – YABUCOA 100 kV is overloaded to 125.5% of its 165 MVA emergency rating for the loss of the JAMESON – FRANK E RATTS STA 100 kV. The project contributes 12.5% DFAX (64.3 MW) to this overload.

For the loss of INTERCESSION CITY PL – N UNION CITY 100 kV line, the N UNION CITY – FRANK E RATTS STATION 100 kV loads to 125.3% of its 137 MVA rating. Project DFAX: 7.0%.

N UNION CITY – SOUTHEAST CHICAGO ENERGY PROJECT 100 kV: 120.9% loading (146 MVA) following the outage of the SOUTHEAST CHICAGO ENERGY PROJECT SUBSTATION – N UNION CITY 100 kV Line 2; DFAX = 7.9%.

For the loss of NORTH UNION CITY – SOUTHEAST CHICAGO ENERGY PROJECT 100 kV Cir. 1, the SOUTHEAST CHICAGO ENERGY PROJECT – NORTH UNION CITY 100 kV Cir. 2 loads to 120.9% of its 146 MVA rating. Project DFAX: 7.9%.

The WATERFLOW – WARRENSBURG 100 kV line is overloaded to 117.4% of its 179 MVA emergency rating for the loss of the WATERFLOW – WARRENSBURG 100 kV #2. The project contributes 9.1% DFAX (47.1 MW) to this overload.

For the loss of WARRENSBURG – WATERFLOW 100 kV Cir. 1, the WATERFLOW SUBSTATION – WARRENSBURG 100 kV Line loads to 117.4% of its 179 MVA rating. Project DFAX: 9.1%.

For the loss of BORDERTOWN – WATERFLOW 100 kV Circuit 2, the BORDERTOWN – WATERFLOW 100 kV Line 1 loads to 116.8% of its 191 MVA rating. Project DFAX: 8.8%.

BORDERTOWN – WATERFLOW 100 kV: 116.8% loading (191 MVA) following the outage of the WATERFLOW – BORDERTOWN 100 kV Circuit 1; DFAX = 8.8%.

For the loss of NORTH UNION CITY – FRANK E RATTS STA 100 kV Line, the N UNION CITY SUBSTATION – INTERCESSION CITY PL 100 kV L loads to 116.3% of its 131 MVA rating. Project DFAX: 6.1%.

The ORTONA – JAMESON 100 kV is overloaded to 113.6% of its 113 MVA emergency rating for the loss of the WARRENSBURG – FLETCHER CHALLENGE 100 kV. The project contributes 9.5% DFAX (48.8 MW) to this overload.

INTERCESSION CITY PL – FRANK E RATTS STATION 100 kV: 112.0% loading (124 MVA) following the outage of the FRANK E RATTS STA – NORTH UNION CITY 100 kV; DFAX = 5.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NACOGDOCHES ROAD SUBSTATION – ELM CRK 100 kV	Rebuild line with higher-capacity conductor	\$1,572,000	63% (\$990,000)
SOUTHEAST CHICAGO ENERGY PROJECT – SHROUD 100 kV line	Rebuild line with higher-capacity conductor	\$3,094,000	63% (\$1,948,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HISKAMP – WILDWOOD 100 kV L	Rebuild line with higher-capacity conductor	\$2,005,000	63% (\$1,263,000)
SHROUD – ELM CRK 100 kV	Rebuild line with higher-capacity conductor	\$1,030,000	63% (\$649,000)
HEGEWISCH – OULOOK 100 kV	Rebuild line with higher-capacity conductor	\$1,249,000	63% (\$787,000)
LAS VEGAS 6306 – WEST MEMPHIS GATEWAY 100 kV	Rebuild line with higher-capacity conductor	\$1,414,000	63% (\$891,000)
HUISKAMP – HEGEWISCH 100 kV	Rebuild line with higher-capacity conductor	\$1,429,000	63% (\$900,000)
FLETCHER CHALLENGE – WARRENSBURG 100 kV	Rebuild line with higher-capacity conductor	\$1,153,000	63% (\$726,000)
NACOGDOCHES RD – LAS VEGAS 6306 100 kV	Rebuild line with higher-capacity conductor	\$2,069,000	63% (\$1,303,000)
TOWER AUTOMOTIVE – HEGEWISCH SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$2,164,000	63% (\$1,362,000)
TOWER AUTOMOTIVE – HEGEWISCH 100 kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,587,000	63% (\$999,000)
JAMESON – FRANK E RATTS STATION 100 kV Line	Rebuild line with higher-capacity conductor	\$2,565,000	63% (\$1,615,000)
W MEMPHIS GATEWAY – HUGO 100 kV	Rebuild line with higher-capacity conductor	\$1,128,000	63% (\$710,000)
HUGO – BLUFFVIEW 100 kV line	Rebuild line with higher-capacity conductor	\$1,273,000	63% (\$802,000)
FLETCHER CHALLENGE – YABUCOA 100 kV	Rebuild line with higher-capacity conductor	\$957,000	63% (\$603,000)
FRANK E RATTS STA – N UNON CITY 100 kV	Rebuild line with higher-capacity conductor	\$598,000	63% (\$376,000)
N UNON CITY – SOUTHEAST CHICAGO ENERGY PROJECT 100 kV	Rebuild line with higher-capacity conductor	\$5,572,000	63% (\$3,509,000)
SOUTHEAST CHICAGO ENERGY PROJECT – NORTH UNION CITY 100 kV #2	Rebuild line with higher-capacity conductor	\$2,728,000	63% (\$1,718,000)
WATERFLOW – WARRENSBURG 100 kV line	Rebuild line with higher-capacity conductor	\$759,000	63% (\$478,000)
WATERFLOW – WARRENSBURG 100 kV	Rebuild line with higher-capacity conductor	\$1,072,000	63% (\$675,000)
BORDERTOWN – WATERFLOW 100 kV Line 1	Rebuild line with higher-capacity conductor	\$1,029,000	63% (\$648,000)
WATERFLOW – BORDERTOWN 100 kV	Rebuild line with higher-capacity conductor	\$933,000	63% (\$588,000)
NORTH UNION CITY – INTERCESSION CITY PL 100 kV	Rebuild line with higher-capacity conductor	\$1,000,000	63% (\$630,000)
JAMESON – ORTONA 100 kV	Rebuild line with higher-capacity conductor	\$2,193,000	63% (\$1,381,000)
INTERCESSION CITY PL SUBSTATION – FRANK E RATTS STATION 100 kV Line	Rebuild line with higher-capacity conductor	\$732,000	63% (\$461,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$26,012,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.